

FCU3

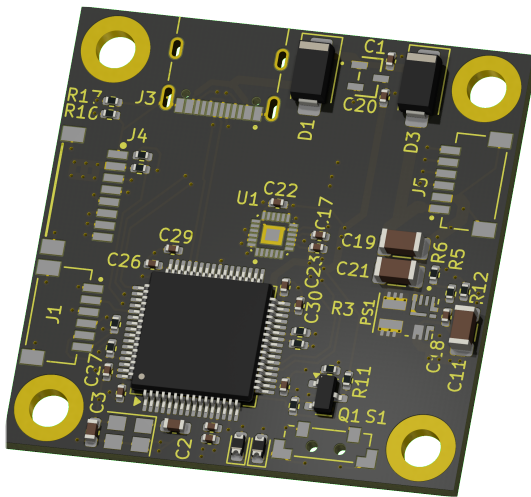
Variant: PRELIMINARY

2025-12-07

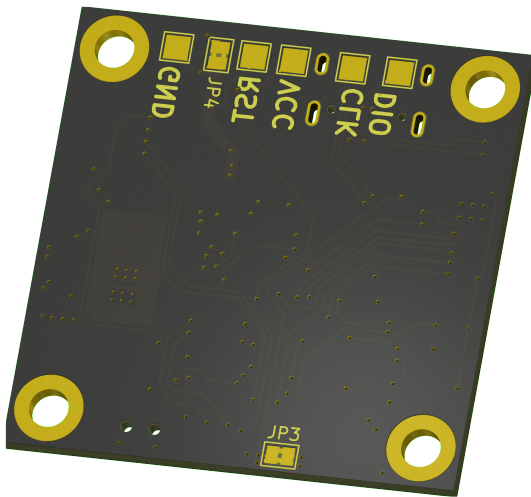
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 07-Dec-2025

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

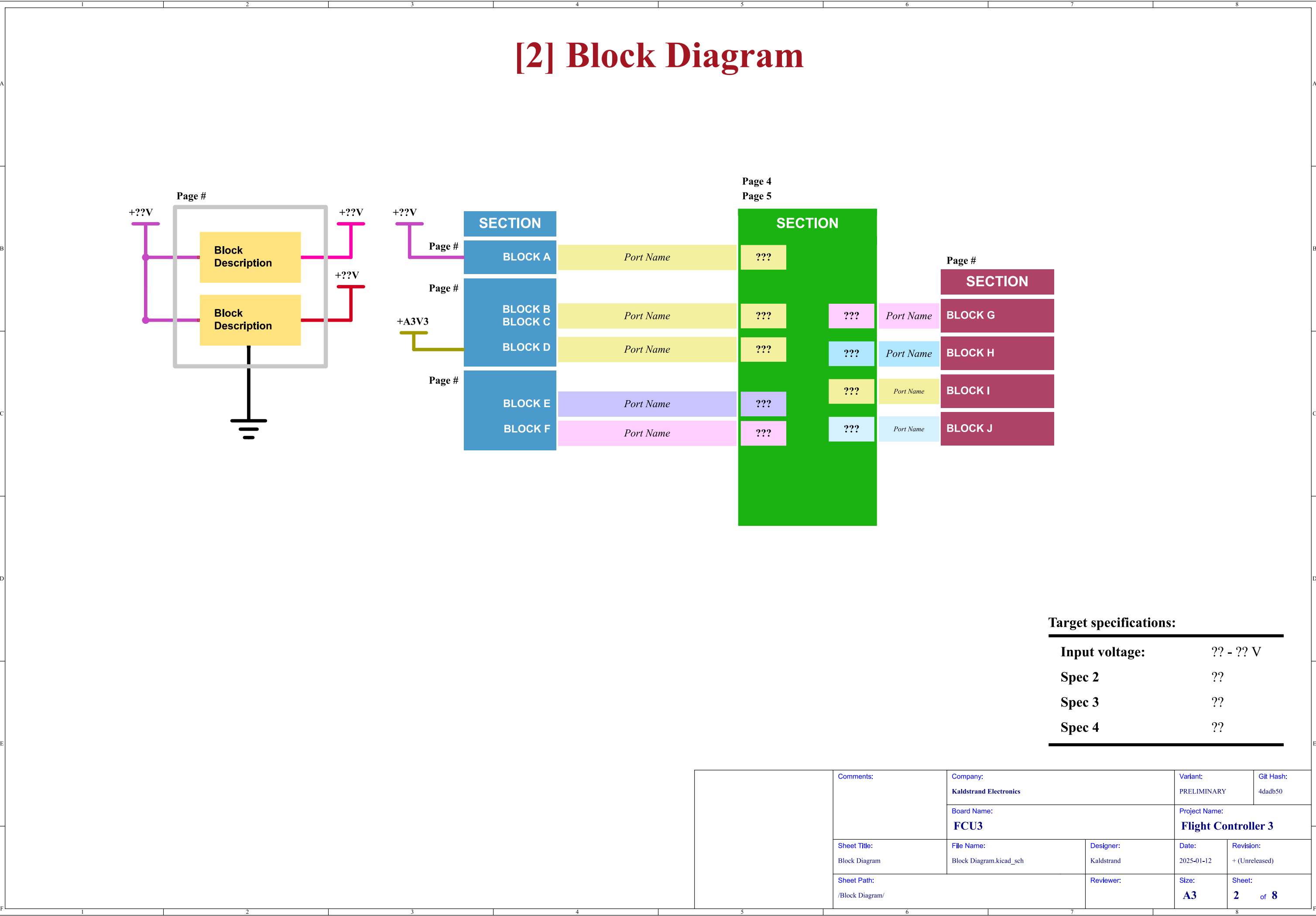
DESIGN NOTE:

Example text for critical design notes.

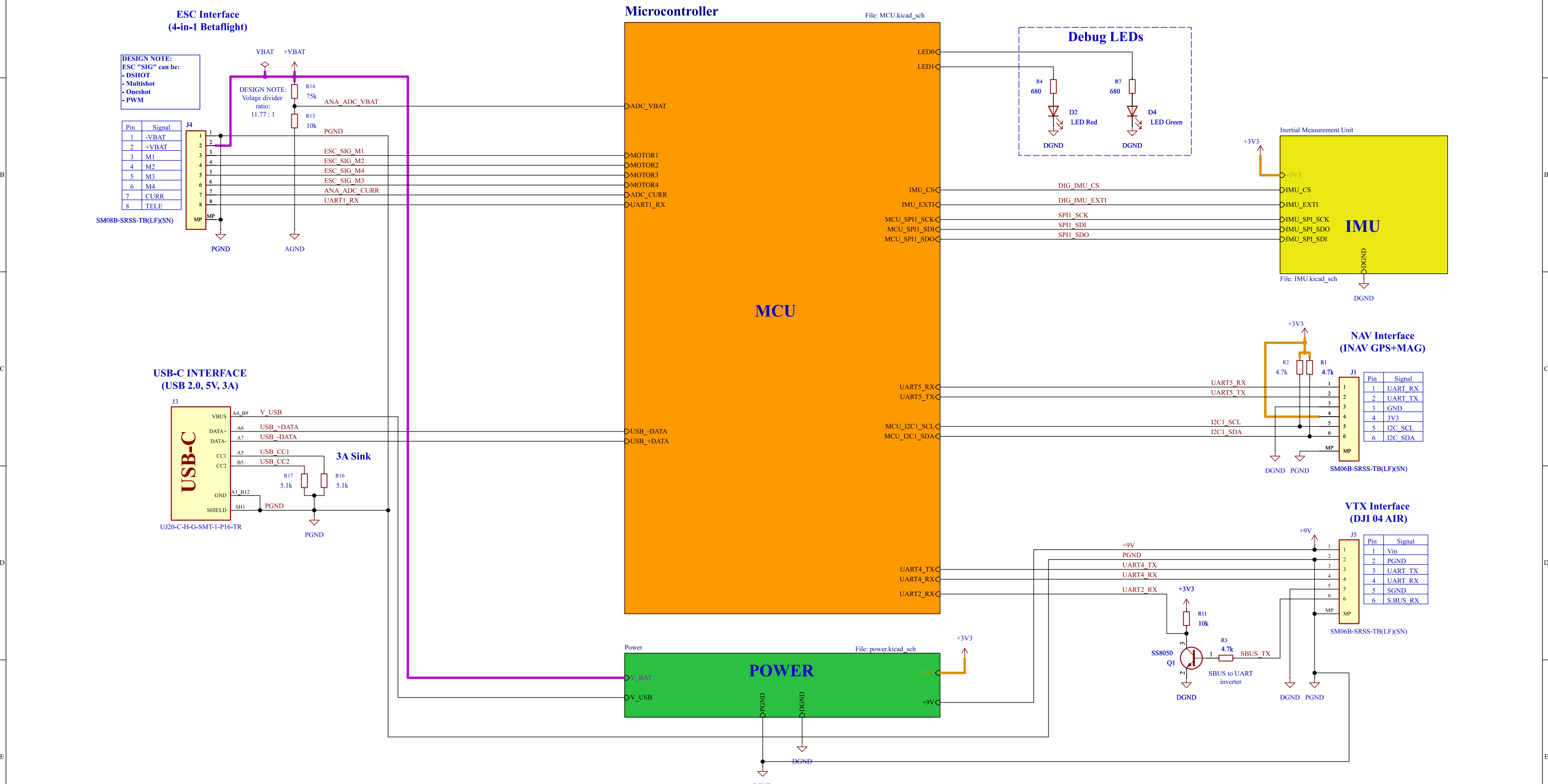
LAYOUT NOTE:

Example text for critical layout guidelines.

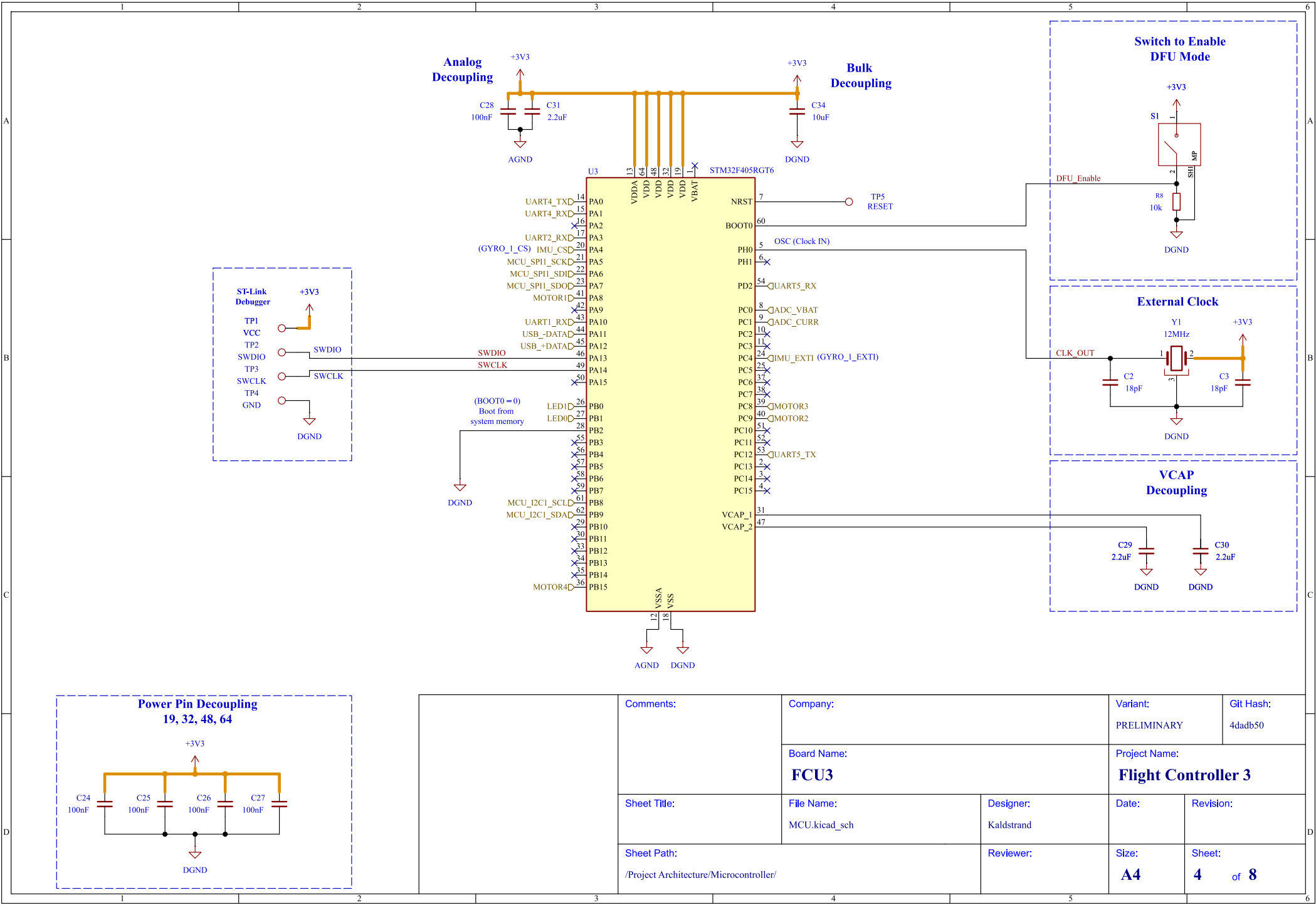
	Comments:	Company: Kaldstrand Electronics	Variant: PRELIMINARY	Git Hash: 4dad50
	Sheet Title:	Board Name: FCU3	Project Name: Flight Controller 3	
	Sheet Path: /	File Name: FCU3.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12
			Revision: + (Unreleased)	
			Size: A3	Sheet: 1 of 8

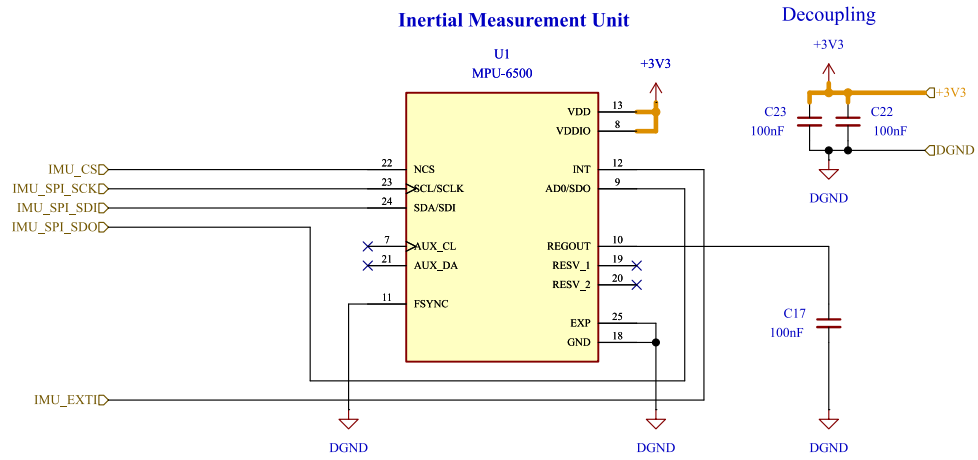


[3] Project Architecture



Grounding Strategy Ground Ties DGND JP3 AGND DGND AGND Mounting Holes H1 H2 H3 H4 PGND										PGND	
		Comments:		Company: Kaldstrand Electronics		Variant: PRELIMINARY		Git Hash: 4dad50			
				Board Name: FCU3		Project Name: Flight Controller 3					
		Sheet Title: Project Architecture		File Name: Project Architecture.kicad_sch		Designer: Kaldstrand		Date: 2025-01-12		Revision: + (Unreleased)	
		Sheet Path: /Project Architecture/				Reviewer:		Size: A3		Sheet: 3 of 8	



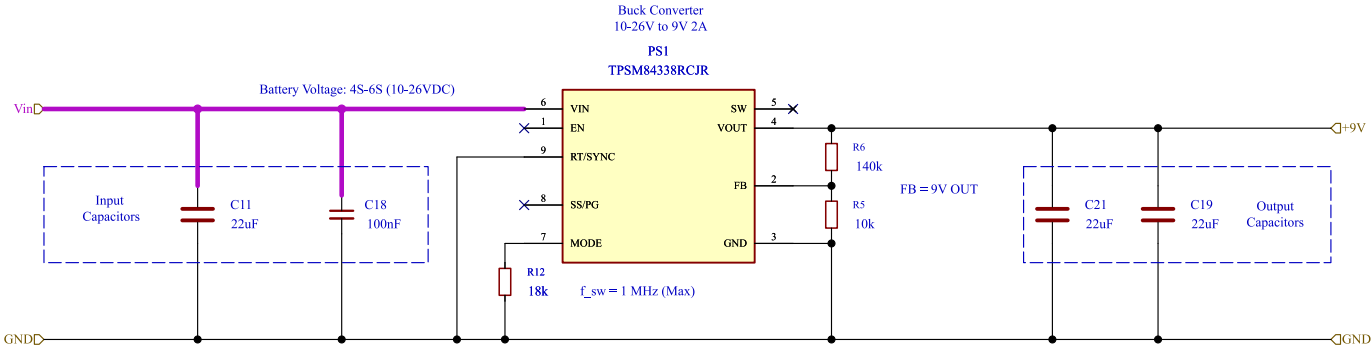


	Comments:	Company:	Variant: PRELIMINARY	Git Hash: 4dad50
		Board Name: FCU3	Project Name: Flight Controller 3	
	Sheet Title:	File Name: IMU.kicad_sch	Designer: Kaldstrand	Date: Revision:
	Sheet Path: /Project Architecture/Inertial Measurement Unit/		Reviewer:	Size: A4 Sheet: 5 of 8

DESIGN NOTE:
 $R_{FBB} = (V_{OUT} - V_{REF}) / V_{REF} * R_{FBB}$
 $R_{FBB} = (9V - 0.6V) / 0.6V * 10k$
 $R_{FBB} = 140k$

 $V_{OUT} = 9V, V_{REF} = 0.6V, R11 = 10k, R6 = 140k$

To-Do:
Add design notes
explaining DC/DC settings.



	Comments:	Company:	Variant: PRELIMINARY	Git Hash: 4dad50
		Board Name: FCU3	Project Name: Flight Controller 3	
	Sheet Title:	File Name: stepdown_10-26vin_9v2a.kicad_sch	Designer: Kaldstrand	Date: Revision:
	Sheet Path: /Project Architecture/Power/10-26V to 9V2A Step-Down Converter/	Reviewer:	Size: A4	Sheet: 8 of 8

[9] Revision History

	Comments:	Company: Kaldstrand Electronics		Variant: PRELIMINARY	Git Hash: 4dad50
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 9 of 8